

STD 1: Statistical Analysis (Std 1), S1 Data Analysis (Y11)

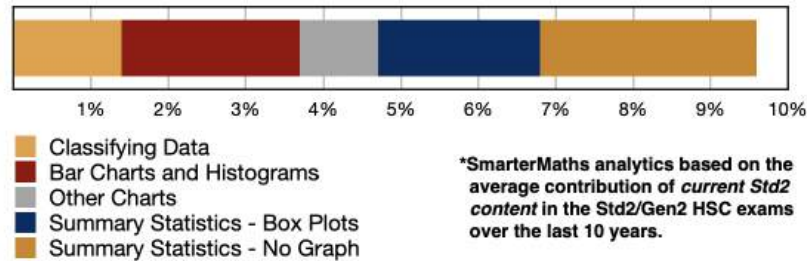
Summary Statistics - Box Plots (Std 1)

Teacher: Denis Vlismas

Exam Equivalent Time: 49.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Std2/Gen2 Exam Contribution History S1 Data Analysis

MS-S1 Data Analysis



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-S1 Data Analysis has accounted for 6% and 5% in the 2020 and 2019 Std1 exams respectively.
- MS-S1 Data Analysis is split into the sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the "relative" importance of each sub-topic.
- This specific analysis looks at the minor sub-topic *Box-plots*.

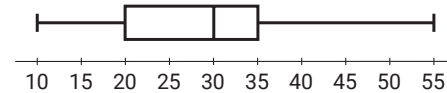
ANALYSIS - What to Expect and Common pitfalls

- Box-plots and in particular, the *comparison of parallel box-plots*, receives its own dot point in the new syllabus, confirming its ongoing importance in the Std1 exam.
- Box-plots were examined in a 1-mark multiple choice question in the 2020 Std1 exam. We note that this topic area has occasionally attracted chunky 4-6 mark allocations, particularly for *parallel box-plot* questions, and must be a revision focus for all students sitting the Std1 exam.

Questions

1. Statistics, STD2 S1 2017 HSC 1 MC

The box-and-whisker plot for a set of data is shown.

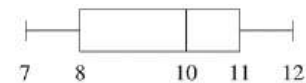


What is the median of this set of data?

- A. 15
- B. 20
- C. 30
- D. 35

2. Statistics, STD2 S1 2004 HSC 12 MC

This box-and-whisker plot represents a set of scores.



What is the interquartile range of this set of scores?

- (A) 1
- (B) 2
- (C) 3
- (D) 5

3. Statistics, STD2 S1 SM-Bank 2 MC

A dataset has the following five-number summary.

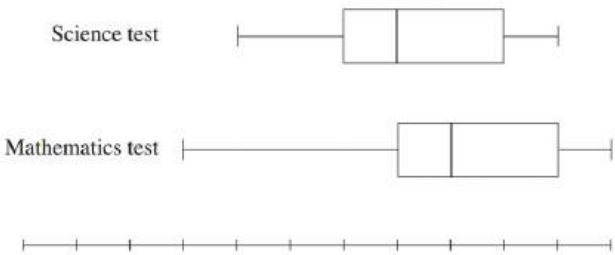
Minimum value	?
First quartile	10
Median	12
Third quartile	13
Maximum value	15

If the range of the dataset is 8, what is the minimum value of the dataset?

- A. 2
- B. 3
- C. 4
- D. 7

4. Statistics, STD2 S1 2008 HSC 10 MC

The marks for a Science test and a Mathematics test are presented in box-and-whisker plots.

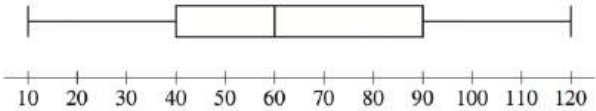


Which measure must be the same for both tests?

- (A) Mean
- (B) Range
- (C) Median
- (D) Interquartile range

5. Statistics, STD2 S1 2015 HSC 6 MC

The times, in minutes, that a large group of students spend on exercise per day are presented in the box-and-whisker plot.

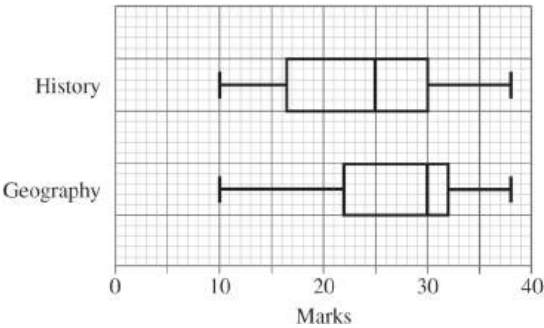


What percentage of these students spend between 40 minutes and 60 minutes per day on exercise?

- (A) 17%
- (B) 20%
- (C) 25%
- (D) 50%

6. Statistics, STD2 S1 2016 HSC 22 MC

The box-and-whisker plots show the results of a History test and a Geography test.



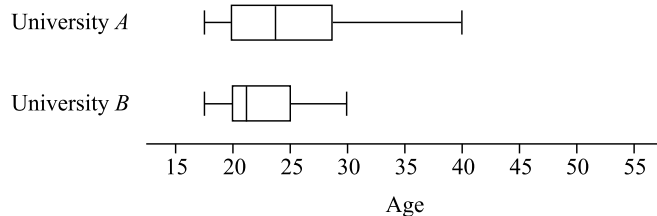
In History, 112 students completed the test. The number of students who scored above 30 marks was the same for the History test and the Geography test.

How many students completed the Geography test?

- (A) 8
- (B) 50
- (C) 56
- (D) 112

7. Statistics, STD1 S1 2020 HSC 2 MC

A random sample of students was taken from each of two universities, and their ages were recorded. The boxplots of their ages are shown.

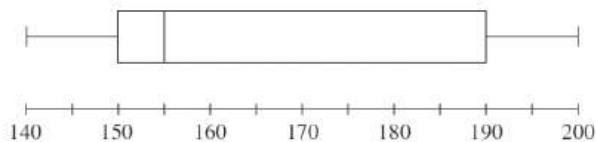


For the given samples of students' ages, which of the following statements is FALSE?

- A.** The range for University A is smaller than the range for University B.
- B.** The median for University A is higher than the median for University B.
- C.** The interquartile range (IQR) for University A is larger than the IQR for University B.
- D.** The oldest student in the sample from University A is older than the oldest student in the sample from University B.

8. Statistics, STD2 S1 2011 HSC 7 MC

A set of data is displayed in this box-and-whisker plot.



Which of the following best describes this set of data?

- (A)** Symmetrical
- (B)** Positively skewed
- (C)** Negatively skewed
- (D)** Normally distributed

9. Statistics, STD2 S1 2016 HSC 19 MC

A soccer referee wrote down the number of goals scored in 9 different games during the season.

2, 3, 3, 3, 5, 5, 8, 9, ...

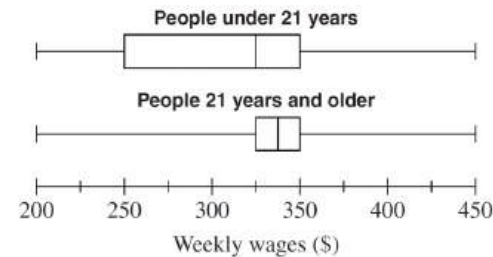
The last number has been omitted. The range of the data is 10.

What is the five-number summary for this data set?

- (A)** 2, 3, 5, 8.5, 12
- (B)** 2, 3, 5, 8.5, 10
- (C)** 2, 3, 5, 8, 12
- (D)** 2, 3, 5, 8, 10

10. Statistics, STD2 S1 2005 HSC 22 MC

Two groups of people were surveyed about their weekly wages. The results are shown in the box-and-whisker plots.



Which of the following statements is true for the people surveyed?

- (A)** The same percentage of people in each group earned more than \$325 per week.
- (B)** Approximately 75% of people under 21 years earned less than \$350 per week.
- (C)** Approximately 75% of people 21 years and older earned more than \$350 per week.
- (D)** Approximately 50% of people in each group earned between \$325 and \$350 per week.

11. Statistics, STD2 S1 SM-Bank 1

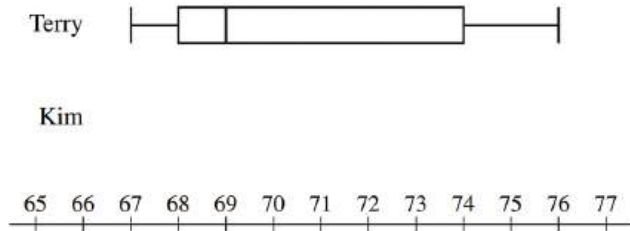
Write down the five-number summary for the dataset 3, 7, 8, 11, 13, 18. **(2 marks)**

12. Statistics, STD2 S1 2014 HSC 29c

Terry and Kim each sat twenty class tests. Terry's results on the tests are displayed in the box-and-whisker plot shown in part (i).

- i. Kim's 5-number summary for the tests is 67, 69, 71, 73, 75.

Draw a box-and-whisker plot to display Kim's results below that of Terry's results. (1 mark)



- ii. What percentage of Terry's results were below 69? (1 mark)

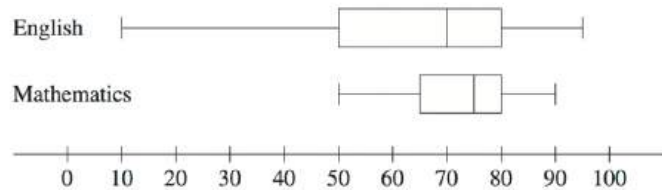
- iii. Terry claims that his results were better than Kim's. Is he correct?

Justify your answer by referring to the summary statistics and the skewness of the distributions.

(4 marks)

13. Statistics, STD2 S1 2012 HSC 28d

The test results in English and Mathematics for a class were recorded and displayed in the box-and-whisker plots.

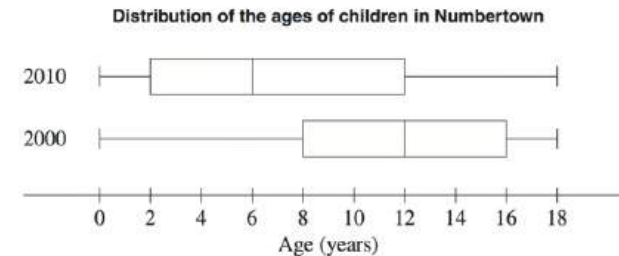


- i. What is the interquartile range for English? (1 mark)

- ii. Compare and contrast the two data sets by referring to the skewness of the distributions and the measures of location and spread. (3 marks)

14. Statistics, STD2 S1 2010 HSC 27b

The graphs show the distribution of the ages of children in Numbertown in 2000 and 2010.



- i. In 2000 there were 1750 children aged 0–18 years.

How many children were aged 12–18 years in 2000? (1 mark)

- ii. The number of children aged 12–18 years is the same in both 2000 and 2010.

How many children aged 0–18 years are there in 2010? (1 mark)

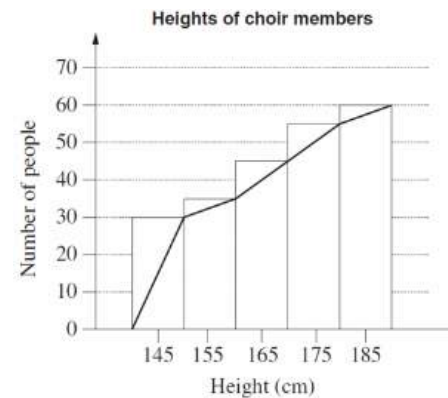
- iii. Identify TWO changes in the distribution of ages between 2000 and 2010. In your answer, refer to measures of location or spread or the shape of the distributions. (2 marks)

- iv. What would be ONE possible implication for government planning, as a consequence of this change in the distribution of ages? (1 mark)

15. Statistics, STD2 S1 2006 HSC 24c

The heights of the 60 members of a choir were recorded. These results were grouped and then displayed as a cumulative frequency histogram and polygon.

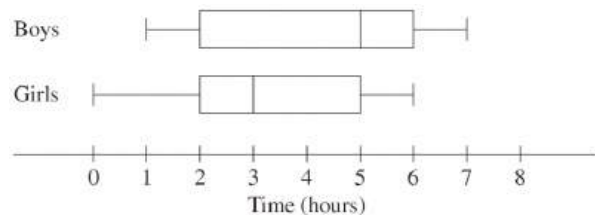
The shortest person in the choir is 140 cm and the tallest is 190 cm.



Draw an accurate box-and-whisker plot to represent the data. (3 marks)

16. Statistics, STD2 S1 2009 HSC 26a

In a school, boys and girls were surveyed about the time they usually spend on the internet over a weekend. These results were displayed in box-and-whisker plots, as shown below.



- Find the interquartile range for boys. **(1 mark)**
- What percentage of girls usually spend 5 or less hours on the internet over a weekend? **(1 mark)**
- Jenny said that the graph shows that the same number of boys as girls usually spend between 5 and 6 hours on the internet over a weekend.
Under what circumstances would this statement be true? **(1 mark)**

Worked Solutions

- Statistics, STD2 S1 2017 HSC 1 MC

$$\begin{aligned}\text{Median} &= 30 \\ \Rightarrow C\end{aligned}$$

- Statistics, STD2 S1 2004 HSC 12 MC

$$\begin{aligned}Q_1 &= 8 \\ Q_3 &= 11 \\ IQR &= Q_3 - Q_1 \\ &= 11 - 8 \\ &= 3 \\ \Rightarrow C\end{aligned}$$

- Statistics, STD2 S1 SM-Bank 2 MC

$$\begin{aligned}\text{Range} &= \text{Max Value} - \text{Min Value} \\ 8 &= 15 - \text{Min Value} \\ \therefore \text{Min Value} &= 15 - 8 \\ &= 7 \\ \Rightarrow D\end{aligned}$$

- Statistics, STD2 S1 2008 HSC 10 MC

$$\begin{aligned}IQR &= \text{Upper Quartile} - \text{Lower Quartile} \\ \Rightarrow D\end{aligned}$$

- Statistics, STD2 S1 2015 HSC 6 MC

$$\begin{aligned}Q_1 &= 40 \\ \text{Median} &= 60 \\ \therefore \% \text{ Students between 40 and 60} \\ &= 50\% - 25\% \\ &= 25\% \\ \Rightarrow C\end{aligned}$$

6. Statistics, STD2 S1 2016 HSC 22 MC

In History

$$Q_3 = 30$$

$\therefore$ Students scoring over 30

$$= 25\% \times 112$$

$$= 28$$

In Geography:

$$\text{Median} = 30$$

$\therefore$ Students that completed Geography

$$= 2 \times 28 = 56$$

$\Rightarrow C$

7. Statistics, STD1 S1 2020 HSC 2 MC

Consider A :

$$\text{Range of } A \approx 40 - 17.5 \approx 22.5$$

$$\text{Range of } B \approx 30 - 17.5 \approx 12.5$$

$\therefore$ Range for $A >$ Range for B

$\Rightarrow A$

8. Statistics, STD2 S1 2011 HSC 7 MC

Since the median (155) is closer to the lower quartile (150) and lower extreme (140) than the upper equivalents, it is positively skewed.

$\Rightarrow B$

♦ Mean mark 47%

9. Statistics, STD2 S1 2016 HSC 19 MC

Since range is 10,

Last data point = 12

$$Q_1 = 3$$

$$Q_3 = \frac{8 + 9}{2} = 8.5$$

Median = 5

$\Rightarrow A$

♦ Mean mark 46%.

10. Statistics, STD2 S1 2005 HSC 22 MC

Considering A

50% of Under 21 group earned over \$325 and 75% of Over 21 group did. NOT TRUE.

Considering B

75% of Under 21 group earned below \$350 is TRUE.

C and D can both be proven to be untrue using their median and quartile values.

$\Rightarrow B$

11. Statistics, STD2 S1 SM-Bank 1

Minimum value: 3

First quartile: 7

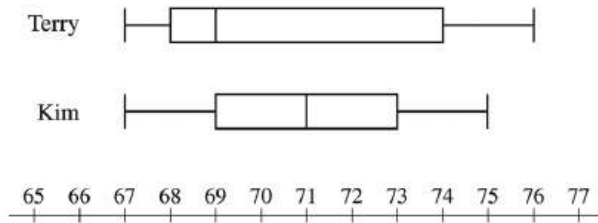
$$\text{Median: } \frac{11 + 8}{2} = 9.5$$

Third quartile: 13

Maximum value: 18

12. Statistics, STD2 S1 2014 HSC 29c

i.



ii. 50%

♦ Mean mark 39%

iii. Terry's results are more positively skewed than

Kim's and also have a higher limit high.

However, Kim's results are more consistent, showing a tighter IQR. They also have a significantly higher median than Terry's and are evenly skewed.

∴ Kim's results were better.

♦♦ Mean mark 29%
COMMENT: Examiners look favourably on using *language of location* in answers, particularly the areas they have specifically pointed students towards (skewness in this example).

13. Statistics, STD2 S1 2012 HSC 28d

i. $IQR_{(\text{English})} = 80 - 50$
 $= 30$

ii. Skewness

- English has greater negative skew
- Maths is more normally distributed

Location and Spread

- English has a range of 85, Maths has 40.
- English has larger IQR than Maths (30 vs 15)
- Maths' Median (75) is higher than English (70)
- Same upper quartile marks (80)
- English has highest and lowest individual mark

♦ Mean mark 35%
MARKER'S COMMENT: Markers are looking for students to use the correct language of location and spread such as *mean, median, interquartile range, standard deviation and skewness*.

14. Statistics, STD2 S1 2010 HSC 27b

i. Since the median = 12 years

⇒ 50% of children are aged 12-18 years

♦ Mean mark 45%

∴ # Children 12-18 = $50\% \times 1750$
 $= 875$

ii. Upper quartile (2010) = 12 years

Children in upper quartile = 875 (from part (i))

♦♦ Mean mark 25%

∴ # Children aged 0-18 = 4×875
 $= 3500$

iii. Changes in distribution (only 2 req'd)

- the lower quartile age is lower in 2010
- the median is lower in 2010
- the upper quartile age is lower in 2010
- the interquartile range is greater in 2010
- 2010 is positively skewed while 2000 is negatively

♦ Mean mark 35%
MARKER'S COMMENT: A number of students incorrectly identified 'positive' skew as 'negative' skew here.

iv. Implication for government planning

- since the children are getting younger in 2010, approve and build more childcare facilities
- Build more school and public playgrounds.

♦ Mean mark 46%
MARKER'S COMMENT: This is worth only 1 mark and long-winded answers are not required.

15. Statistics, STD2 S1 2006 HSC 24c

Low = 140

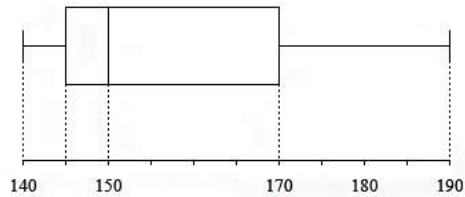
High = 190

Median = 150 (# People = 30)

$Q_1 = 145$ (# People = 15)

$Q_3 = 170$ (# People = 45)

Box and Whisker



16. Statistics, STD2 S1 2009 HSC 26a

i. Interquartile range = $6 - 2$

$$= 4$$

ii. Upper quartile = 5

$\therefore$ 75% of girls spend 5 or less hours

♦♦ Mean mark part ii: 31%

♦♦♦ Mean mark part iii: 9%

iii. 5-6 hours for girls accounts for 25% of all girls.

5-6 hours for boys accounts for 25% of all boys,

(median to the upper quartile represents 25%.)

$\Rightarrow$ This will only be the same number if the number of all girls surveyed equals the number of boys surveyed.